SEAN KHOMPHENGCHAN

CURRENT ADDRESS
2176 Lucia Lane
West Lafayette, IN 47906

Active Secret Clearance
574-327-9116
skhomphe@purdue.edu

SOCIALS
seankhom.github.io
linkedin.com/in/seankhom

OBJECTIVE

To obtain an internship or cooperative education opportunity in a Guidance, Navigation, & Controls or Software related position applicable to the aerospace industry.

EDUCATION

Purdue University, West Lafayette, IN — December 2023

Bachelor of Science in Aeronautical & Astronautical Engineering
Specializations: Dynamics and Control
Minor: Astronomy
GPA: 3.19 / 4.00
Relevant Coursework: Dynamics, Multi Agent Autonomy, Control Systems, and Satellite Navigation and Positioning
Technical Skills: MATLAB, Simulink, Python, C, C++, Siemens NX, Kalman Filtering, Git, Agile Development

TECHNICAL & RELEVANT EXPERIENCE

Simulation and Modeling Architecture Systems Engineering Intern — June 2022 – Present

Raytheon Missiles and Defense, Arlington VA

- Developed algorithms to predict and simulate solutions in systems and to conduct threat assessments
- Briefed executives all the way to the president of the company on work accomplished and impact of.
- Worked in a team and independently to successfully integrate subsystems into an overarching structure
- Utilized Object Oriented Python and Java to build campaign level simulation software

Undergraduate Teaching Assistant for the First Year Engineering Program — August 2021 – Present

Purdue University, West Lafayette IN

- Provide guidance to over 100 students on MATLAB, Python, Excel to refine course concepts
- Mentor and guide teams of students through their projects, and navigation of their academic careers
- Run help sessions to provide personalized instruction to individual students and small groups
- Facilitate handling of supplies and materials for class and other organizational duties

Undergraduate Research Assistant at Ma Quantum Lab — January 2021 – June 2021

Purdue University, West Lafayette IN

- Created and presented a poster using Latex at the Purdue Undergraduate Research Conference
- Researched Microwave electronics for superconducting quantum circuits for use on quantum simulation testing
- Realized noise reduction of 90% on a log curve for when trying to analyze signal received from system
- Collaborated with Principal Investigator and lab group to exchange idea for finding better methods for end to end testing of system

PROJECTS

Project ASTRA — January 2022 – Present

- Utilized Kalman filtering and other filtration methods to predict and verify sensor data accuracy
- Create and model motion of spacecraft using sensor data with MATLAB and Simulink
- Designed a rocket that can be controlled midair with PSP Active Controls group

Automatic Inventory and Shipment Management Tool — June 2021 – July 2021

- Developed program for Elkhart Brass to manage and track status of parts for production
- Scripted in Python 3 with the Pandas, Numpy, and Pyautogui Libraries
- Saved client 13 hours of time so time may be spent on other facets of production